The Effect of Information Entropy on Recognition Memory for Words

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Background & Research Question

Background

- Language comprehension is predictive.
- Predictable words can lead to false memories (Hubbard et al., 2019).
- Unexpected words often receive deeper semantic processing (Rommers & Federmeier, 2018).

Problem

Traditional measures (e.g., cloze probability) only capture the most likely completion and do not reflect the *distribution* of predicted alternatives.

Information-Theoretic Perspective

Prediction uncertainty can be quantified using Shannon entropy. $H(X) = -\sum_{i=1}^{n} p_i \log_2(p_i)$

Entropy may have two distinct components (Karimi, Zinn, & Li, under review). More (equi-probable responses) may correspond to greater:

- Semantic overlap – semantic future overlap between the target word and other responses
- Lexical entropy – competition among lexical candidates

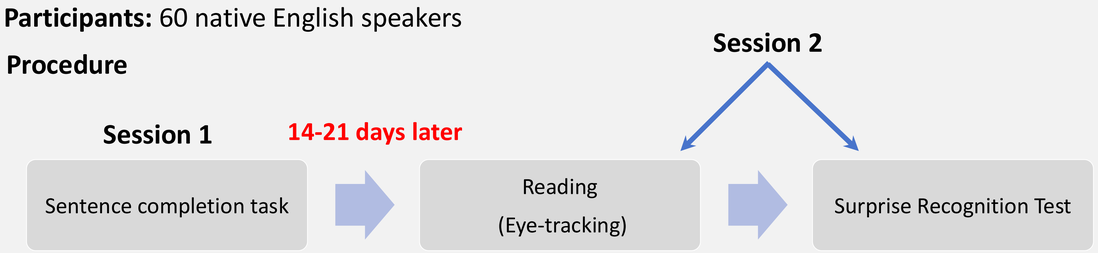
Research Question

Do semantic overlap and lexical entropy influence subsequent recognition memory for words?

Method

Participants: 60 native English speakers

Procedure



In polluted cities, you can barely breathe the ...

	Word	%
word1	air	90
word2	smoke	80
word3	oxygen	70
word4	scent	40
word5	smog	70
word6		
word7		
word8		

a) Low Constraint	They suspected there was something wrong with the air outside the building at noon.
b) Medium Constraint	The factory built expensive machines to filter the air outside the building at noon.
c) High Constraint	In polluted cities, you can barely breathe the air outside the building at noon.

air

Old New

Results & Discussion

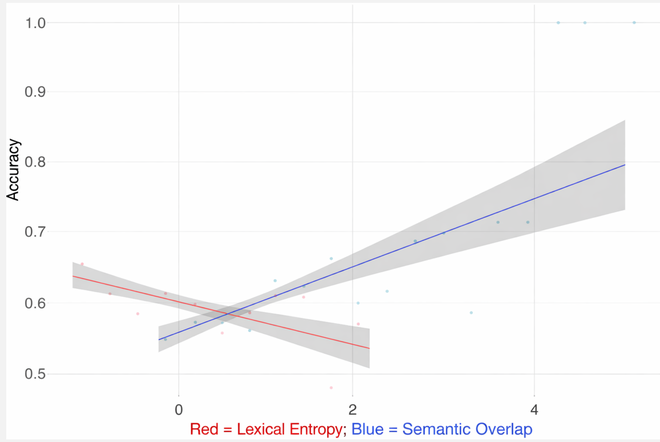
Results

- Mixed-effects logistic regression results:

Predictor	Estimate	SE	z	p
Intercept	0.55	0.13	4.22	< .001
Semantic overlap	0.20	0.03	6.19	< .001
Lexical entropy	-0.09	0.03	-3.04	.002
Reading time	0.04	0.03	1.59	.112

Semantic Overlap
(z = 6.19, p < .001)
Semantic overlap ↑ → Memory accuracy ↑

Lexical Entropy
(z = –3.04, p = .002)
Lexical entropy ↑ → Memory accuracy ↓



Discussion

Prediction structure influences memory through distinct mechanisms.

- Semantic overlap → richer semantic cues → facilitates retrieval
- Lexical entropy → stronger lexical competition → impairs retrieval

Prediction therefore shapes not only online language processing but also memory encoding and retrieval.

Main References

Hubbard, R. J., Rommers, J., Jacobs, C. L., & Federmeier, K. D. (2019). Downstream behavioral and electrophysiological consequences of word prediction on recognition memory. *Frontiers in human neuroscience*, *13*, 291.

Karimi, H., Zinn, J., & Li, H. (under review). Entropy's dilemma: Facilitatory and inhibitory effects of information entropy during language processing.

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